

1650 ▶ 1670



Von Guericke made an early friction machine for producing static electricity.

Von Guericke carries out his experiment at Magdeburg, Germany

1654

Display of power

To demonstrate his understanding of vacuums, German inventor Otto von Guericke made two copper hemispheres, sealed them together, and emptied them of air using a vacuum pump he had invented earlier. Then he got two teams of eight horses to try and pull the hemispheres apart. They failed to do so because the external air pressure that kept the hemispheres pressed together so tightly was too strong.

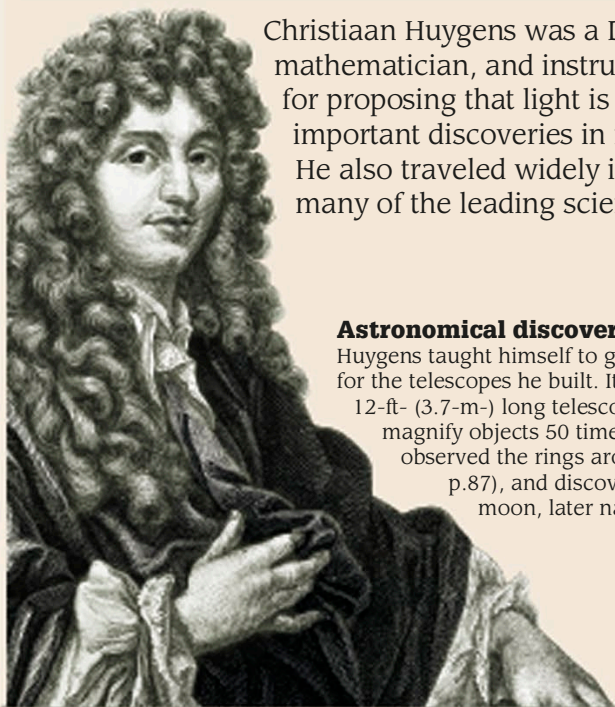


1650

1655

1629–1695

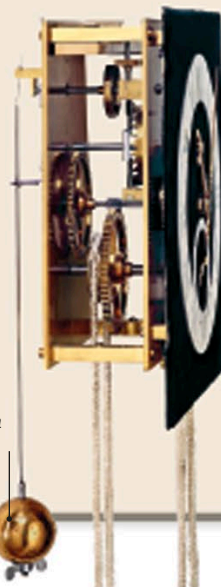
CHRISTIAAN HUYGENS



Christiaan Huygens was a Dutch physicist, astronomer, mathematician, and instrument-maker. Best known for proposing that light is made of waves, he made important discoveries in many areas of science. He also traveled widely in Europe and met with many of the leading scientists of the day.

Astronomical discoveries

Huygens taught himself to grind the lenses for the telescopes he built. It was with his 12-ft- (3.7-m-) long telescope, which could magnify objects 50 times, that he first observed the rings around Saturn (see p.87), and discovered its largest moon, later named Titan.



Swinging pendulum keeps regular time.

KEY DATES

1654 Huygens began making improvements to telescopes with his brother Constantijn.

1656 He produced his design for a pendulum clock.

1678 He proposed that light is made up of waves.

1689 He met Isaac Newton on a visit to England.

Pendulum clock

Huygens designed a clock with a weighted pendulum that kept time accurately to within a few seconds a day—a vast improvement on existing clocks.